Threat Intelligence Integration in SIEM

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1. Abstract

This project demonstrates how threat intelligence can be integrated into a Security Information and Event Management SIEM platform aka Splunk to improve detection of malicious activity. Indicators of Compromise IOCs were collected from AlienVault OTX using an automated Python script then cleaned and enriched by assigning severity levels to each indicator. The resulting IOC datasets covering IPv4 addresses, domains, URLs, and file hashes were imported into Splunk as lookup tables. To test the detection capability in a controlled lab environment a second Python script generated realistic simulated security and network logs containing a mix of benign events and intentional IOC matches. Multiple correlation searches were created in Splunk to compare incoming log fields against the IOC lookups and trigger alerts when matches were found for example outbound connections to malicious IPs/domains, access to suspicious URLs, or presence of known malicious file hashes. The results confirm that Splunk lookups and correlation rules can successfully identify threat-intelligence-based indicators within event data and generate actionable alerts demonstrating a practical beginner friendly workflow for threat intelligence integration, validation, and alerting in SIEM.

2. Introduction

Organizations generate a very large amount of security-related data every day. Manually reviewing these logs is difficult because the volume is high and important attack signals can be hidden inside normal harmless traffic for this reason many organizations use a Security Information and Event Management (SIEM) system.

A SIEM is a platform that collects logs from different systems stores them in one place and helps security analysts search, correlate and detect suspicious activity. Instead of checking each device separately the SIEM provides centralized visibility and supports automated detection through searches, dashboards and alerts.

However a SIEM becomes much more effective when it is combined with Threat Intelligence. Threat intelligence is information about known or suspected cyber threats such as Indicators of Compromise (IOCs). IOCs are observable values that may be linked to malicious activity.

By integrating threat intelligence into the SIEM it becomes possible to automatically detect when internal logs contain communication with known malicious infrastructure or when a malicious file indicator appears in endpoint data. This reduces investigation time and helps security teams respond faster.

3. Project Objective

The main objective of this project is to design and implement a complete beginner friendly workflow for Threat Intelligence Integration in Splunk SIEM including:

1. Collecting threat intelligence IOCs from AlienVault OTX.
2. Processing and enriching the IOC data (including assigning severity levels).
3. Importing the IOC lists into Splunk as lookup tables.
4. Ingesting simulated security logs into Splunk.
5. Building correlation searches (detection rules) that match log events against the IOC lookups.
6. Triggering and verifying alerts when IOC matches occur.

4. Threat Intelligence Collection from AlienVault OTX and AI-Assisted Refinement

A. Gathering IOCs from AlienVault OTX (Threat Intelligence Source)

To obtain real threat intelligence data for this project we used AlienVault OTX (Open Threat Exchange) as the external source of Indicators of Compromise (IOCs). OTX is a community-driven threat intelligence platform where security researchers publish collections of threat data called Pulses. Each pulse may contain indicators such as malicious IP addresses, suspicious domains, URLs, and file hashes, often linked to a specific campaign or attack type like (phishing, brute force, scanning activity).

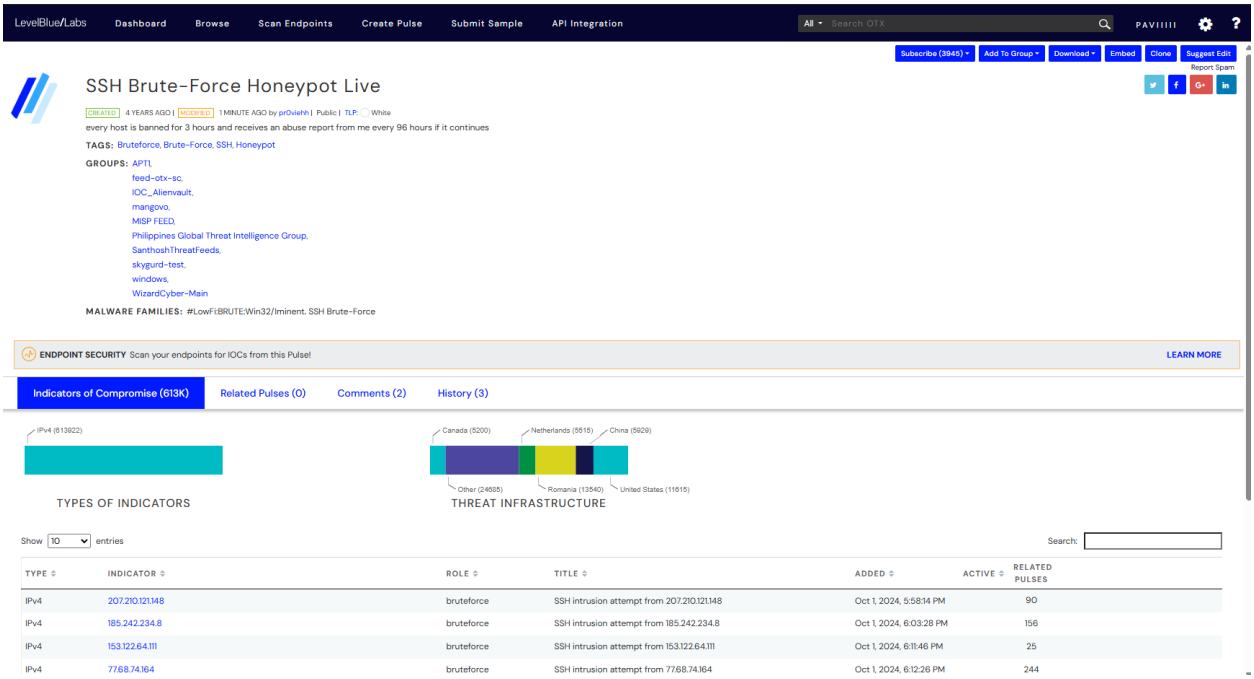


Figure 1- OTX Alien IOC example

In our implementation, we gathered IOCs from OTX by:

1. Selecting relevant pulses based on the project scope network threats such as phishing.
2. Extracting the main IOC attributes:
    - **indicator** (the IOC value, an IP or domain).
    - **indicator type** (IPv4, domain, URL, hash).
    - **pulse/category label** (used later to understand the context of the threat source).
    - **Severity** enrichment field for categorizing later on.

The output of this step was an IOC dataset saved as a CSV file, which later became the input for enrichment, filtering, and Splunk lookup integration.

	A	B	C	D
1	**Indicator type**	**Indicator**	**Type**	**Severity**
2	IPv4	103.77.246.109	SSH Brute-force	Medium
3	IPv4	103.77.246.109	SSH Brute-force	Medium
4	IPv4	103.77.246.109	SSH Brute-force	Medium
5	IPv4	193.46.255.217	SSH Brute-force	Medium
6	IPv4	193.46.255.103	SSH Brute-force	Medium
7	IPv4	193.46.255.159	SSH Brute-force	Medium
8	IPv4	101.126.70.177	SSH Brute-force	Medium
9	IPv4	152.42.235.138	SSH Brute-force	Medium
10	IPv4	59.8.105.70	SSH Brute-force	Medium
11	IPv4	81.192.46.36	SSH Brute-force	Medium
12	IPv4	36.66.16.233	SSH Brute-force	Medium
13	IPv4	87.236.38.246	SSH Brute-force	Medium
14	IPv4	45.148.10.240	SSH Brute-force	Medium
15	IPv4	197.5.145.8	SSH Brute-force	Medium
16	IPv4	61.171.118.216	SSH Brute-force	Medium
17	IPv4	189.178.106.189	SSH Brute-force	Medium
18	IPv4	168.167.228.74	SSH Brute-force	Medium
19	IPv4	101.126.35.9	SSH Brute-force	Medium
20	IPv4	68.183.88.186	SSH Brute-force	Medium

Figure 2 - IOC example after enrichment

B. AI-Assisted Threat Intelligence Enhancement (Finding "Similar" IOCs)

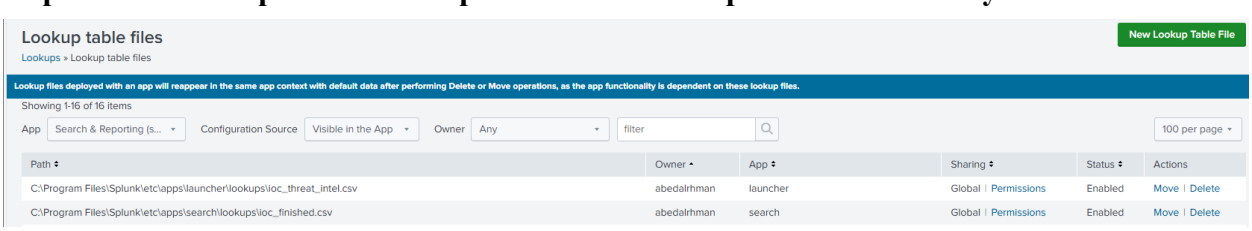
After cleaning the raw IOC data, we performed an additional enhancement step using an AI-assisted approach. The goal was not to replace threat intelligence but to extend it by identifying IOCs that are related to known malicious indicators.

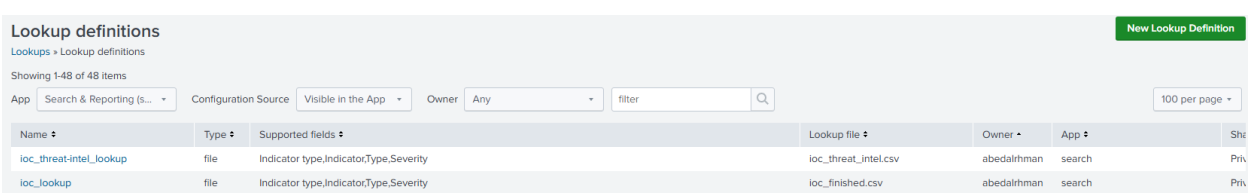
- **Domains:** Attackers often register look-alike domains or create many subdomains under the same base domain.
  AI-assisted logic can help detect:
  - domains sharing the same registrable/base domain,
  - domains with small spelling changes (micros0ft.com vs microsoft.com)
  - suspicious patterns such as random characters or misleading keywords.
- **URLs:** Similarity can be detected by extracting:
  - the domain part
  - repeated path structures (/login/verify, /update/account),
  - or common phishing keywords.
- **IP addresses:** Attack infrastructure is often hosted in the same range.
  A practical "similarity" method is grouping IPs by:
  - the same **/24 subnet** (e.g., x.x.x.*),
  - the same ASN/provider (if enrichment is available).
- **File hashes:** Hashes (MD5).

This AI-assisted refinement strengthens the project by showing an important real-world concept threat intelligence is not only about importing IOCs, but also about improving their quality and using smart methods to discover related malicious infrastructure.

Info	IOC-Threat intelligence	IOC-Threat intelligence – Ai enhanced
Total IOCs	42815	56675
IPv4	39970	50842
Hashes	300	600
Domain	1511	3119
Hostname	948	1942
URL	86	172

Import both into Splunk as lookup tables then look up definitions so they can be indexed.





5. Simulated Log Generation (How the Logs Were Created)

We generated a simulated security log dataset that can be safely ingested into Splunk and used for correlation testing. The logs were created using our Python log generator script which uses our threat-intelligence IOC lists as input and then produces realistic events that look like traffic collected from common security controls.

A. Where the logs come from

The generated dataset simulates logs from multiple typical SOC sources, including:

- ✓ **Firewall** (network connection allow/deny events)
- ✓ **IDS** (intrusion detection alerts/signatures)
- ✓ **DNS** (domain lookup events)
- ✓ **Proxy** (web access and URL requests)
- ✓ **Endpoint/EDR** (host-based detection, file hash visibility)

Each event also includes a Device Information field to represent which tool produced the log (e.g., firewall device, IDS sensor, DNS server, proxy server, endpoint agent).

B. What fields exist in the logs (log structure)

Each generated log record contains practical fields that Splunk can search and correlate, such as:

- ✓ **Timestamp**
- ✓ **Source IP Address / Destination IP Address**
- ✓ **Source Port / Destination Port**
- ✓ **Protocol** (TCP/UDP/ICMP)
- ✓ **Packet Length / Packet Type**
- ✓ **Traffic Type** (Inbound/Outbound)
- ✓ **Payload Data** (short text to mimic content/metadata)
- ✓ **Alerts/Warnings** (used to mark IOC matches)
- ✓ **Attack Type** and **Attack Signature** (filled for malicious events)
- ✓ **Action Taken** (Allowed, Blocked, Allowed (monitored), etc.)
- ✓ **Severity Level** (Info/Medium/High/Critical)
- ✓ **Log Source** (firewall / ids / dns / proxy / endpoint)

C. How we intentionally injected malicious (IOC-matching) logs

To make sure the SIEM rules can be tested properly, the generator does **not** create only benign traffic. Instead, it intentionally embeds IOC values into a controlled portion of the dataset:

- **Total generated logs:** 1250
- **Benign logs:** 800 (Severity = *Info*, no IOC match)
- **Malicious / IOC-matching logs:** 450 (contain explicit IOC indicators and attack context)

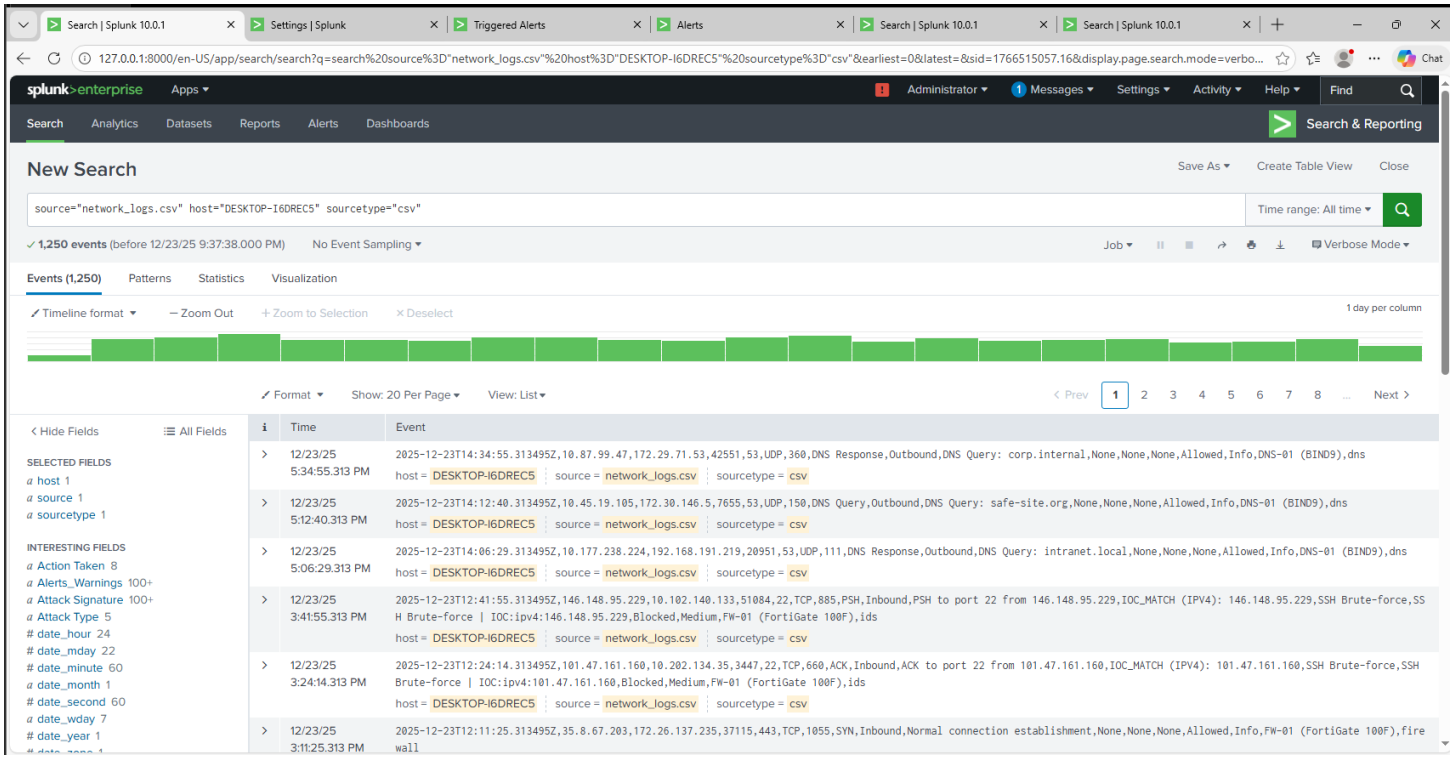


Figure 5 logs in Splunk

D. Number of malicious logs by IOC type

Typer	Number of logs
IPv4	235
Hashes	62
Domain	80
Hostname	30
URL	43
Total	450

This distribution ensures a meaningful set of detections across all required IOC categories (IP, domain, URL, hash), and it provides enough events per category to validate separate correlation rules in Splunk.

6. Building Correlation Searches (Detection Rules) Using IOC Lookups

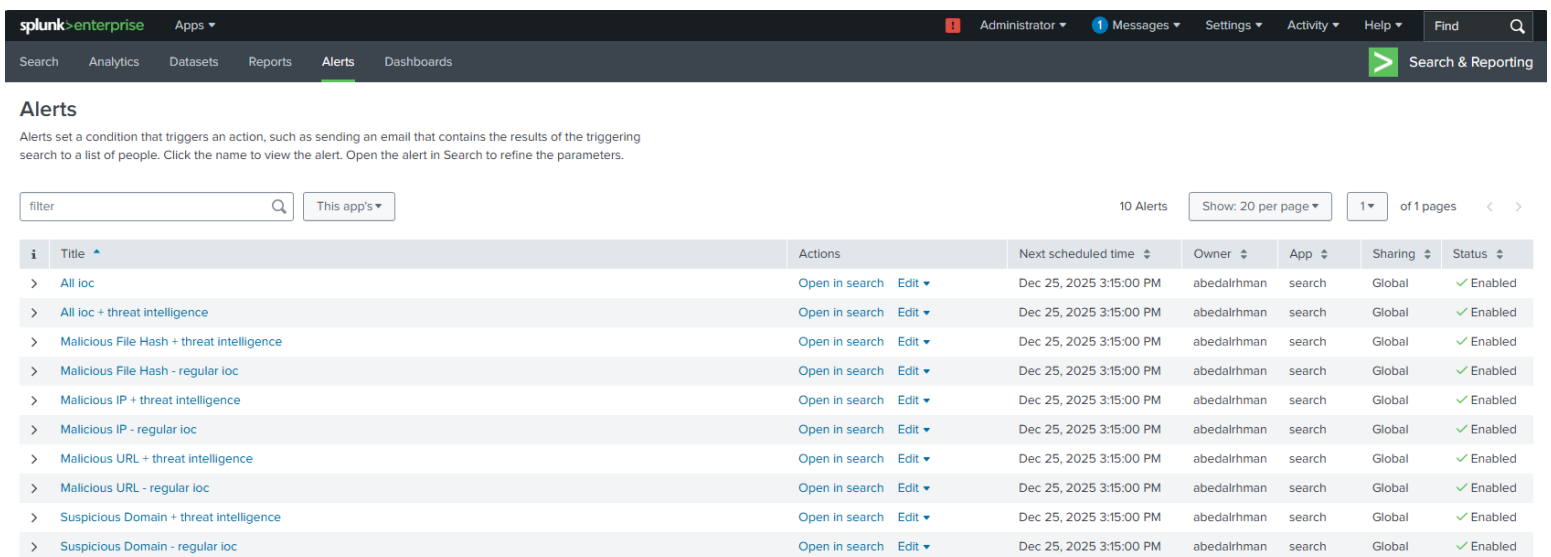
After the IOC datasets were prepared and imported into Splunk as lookup tables we built correlation searches (saved searches/alerts) to detect when any event in our ingested logs matches a known IOC. The goal is simple if a log contains a suspicious IP/domain/URL/hash that exists in the IOC lookup, Splunk should raise an alert.

A. Correlation design (how the rules work)
Each correlation rule follows the same idea:

1. **Take events from the ingested logs** (our simulated dataset).
2. **Extract or use the correct field** depending on IOC type.
3. **Compare that field against the IOC lookup** (regular vs AI-enhanced).
4. **Return matching events** and trigger an alert when matches exist (e.g., results > 0).

B. Rule families we implemented
We created **two versions** of each detection rule:

1. **Regular IOC rule (baseline):** uses the original/regular IOC lookup.
2. **AI-enhanced IOC rule (threat-intelligence enriched):** uses the larger IOC lookup that was expanded using threat intelligence gathering + AI-assisted refinement.



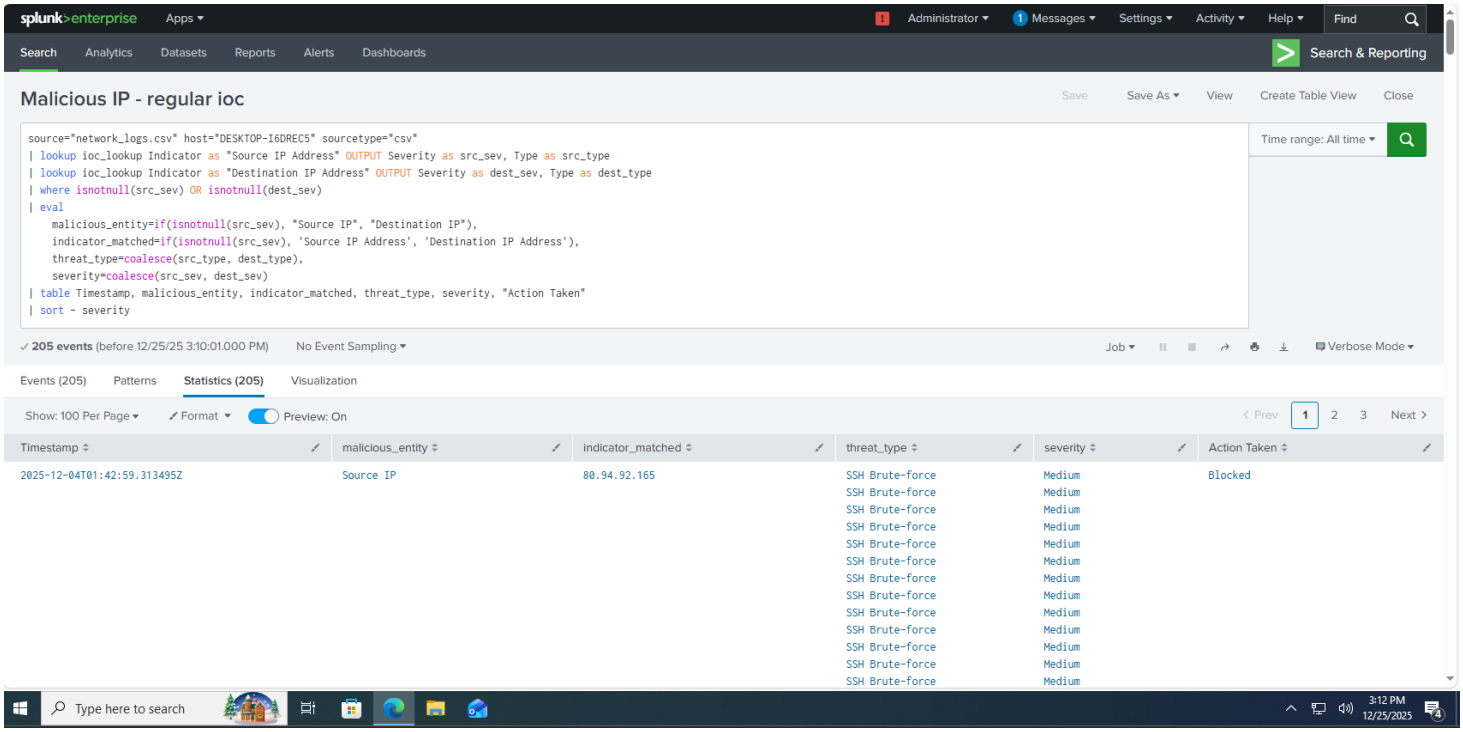
Title	Actions	Next scheduled time	Owner	App	Sharing	Status
All ioc	Open in search Edit	Dec 25, 2025 3:15:00 PM	abedalrhman	search	Global	Enabled
All ioc + threat intelligence	Open in search Edit	Dec 25, 2025 3:15:00 PM	abedalrhman	search	Global	Enabled
Malicious File Hash + threat intelligence	Open in search Edit	Dec 25, 2025 3:15:00 PM	abedalrhman	search	Global	Enabled
Malicious File Hash - regular ioc	Open in search Edit	Dec 25, 2025 3:15:00 PM	abedalrhman	search	Global	Enabled
Malicious IP + threat intelligence	Open in search Edit	Dec 25, 2025 3:15:00 PM	abedalrhman	search	Global	Enabled
Malicious IP - regular ioc	Open in search Edit	Dec 25, 2025 3:15:00 PM	abedalrhman	search	Global	Enabled
Malicious URL + threat intelligence	Open in search Edit	Dec 25, 2025 3:15:00 PM	abedalrhman	search	Global	Enabled
Malicious URL - regular ioc	Open in search Edit	Dec 25, 2025 3:15:00 PM	abedalrhman	search	Global	Enabled
Suspicious Domain + threat intelligence	Open in search Edit	Dec 25, 2025 3:15:00 PM	abedalrhman	search	Global	Enabled
Suspicious Domain - regular ioc	Open in search Edit	Dec 25, 2025 3:15:00 PM	abedalrhman	search	Global	Enabled

Figure 6 All the correlation rules

C. What fields were matched in the logs

To ensure realistic detection, each IOC type was matched to the most relevant log content:

- **IPv4 rules:** compared against **Destination IP Address** (outbound traffic to suspicious servers).



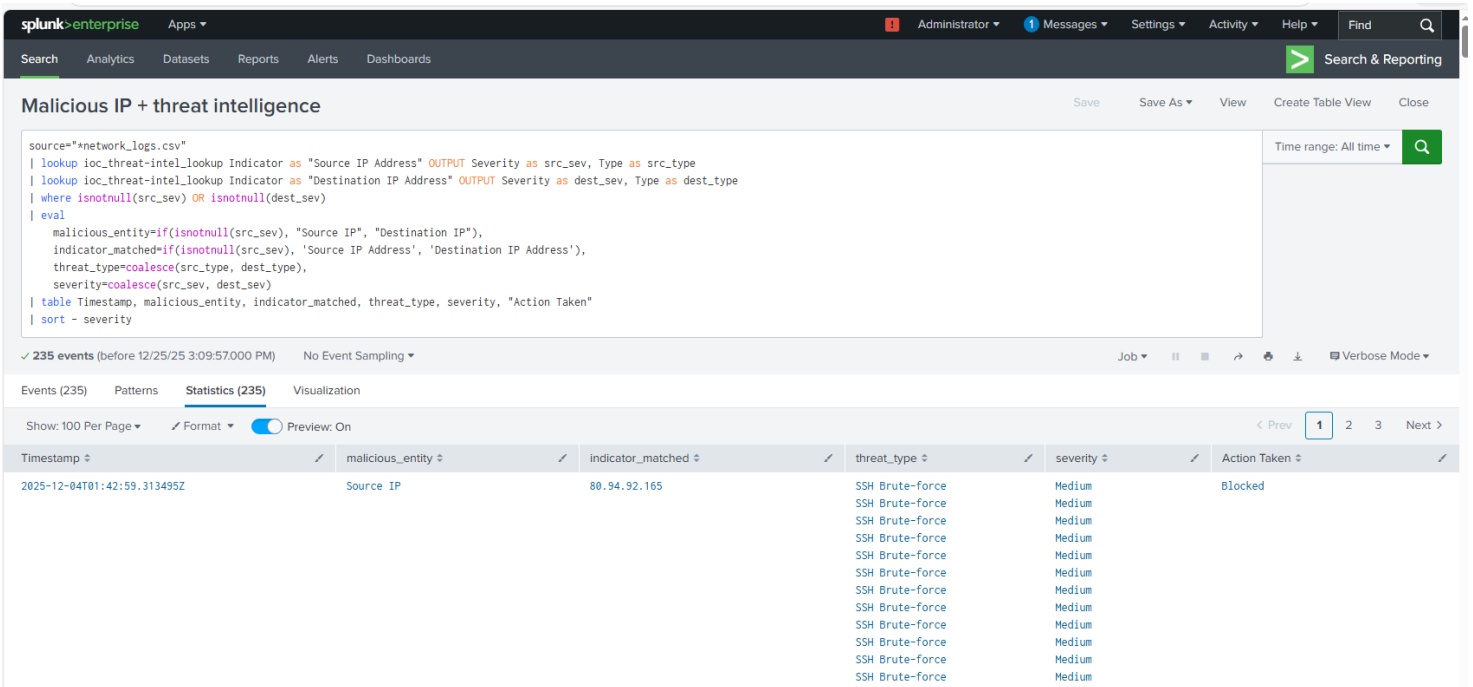
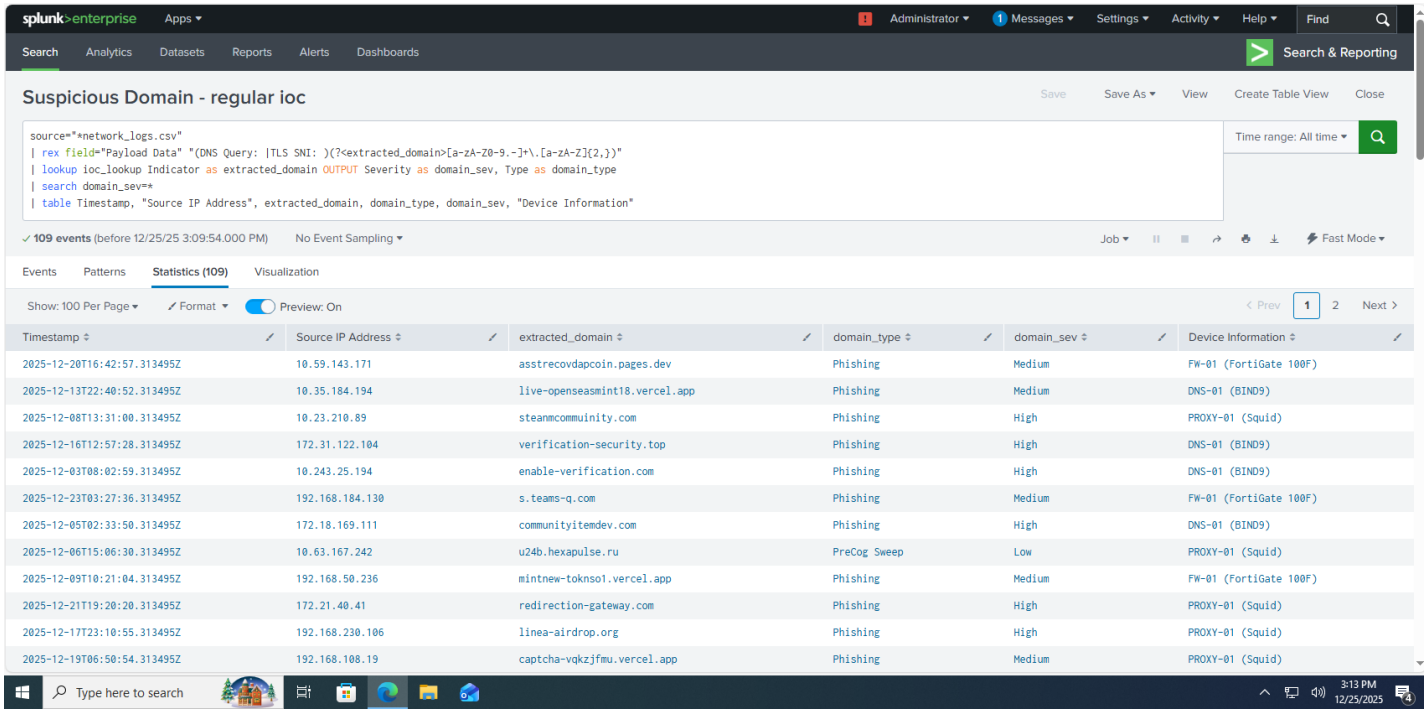


Figure 7 IP correlation rules

- **Domain/Hostname rules:** matched domains/hostnames appearing in the event context (DNS/proxy-related content).



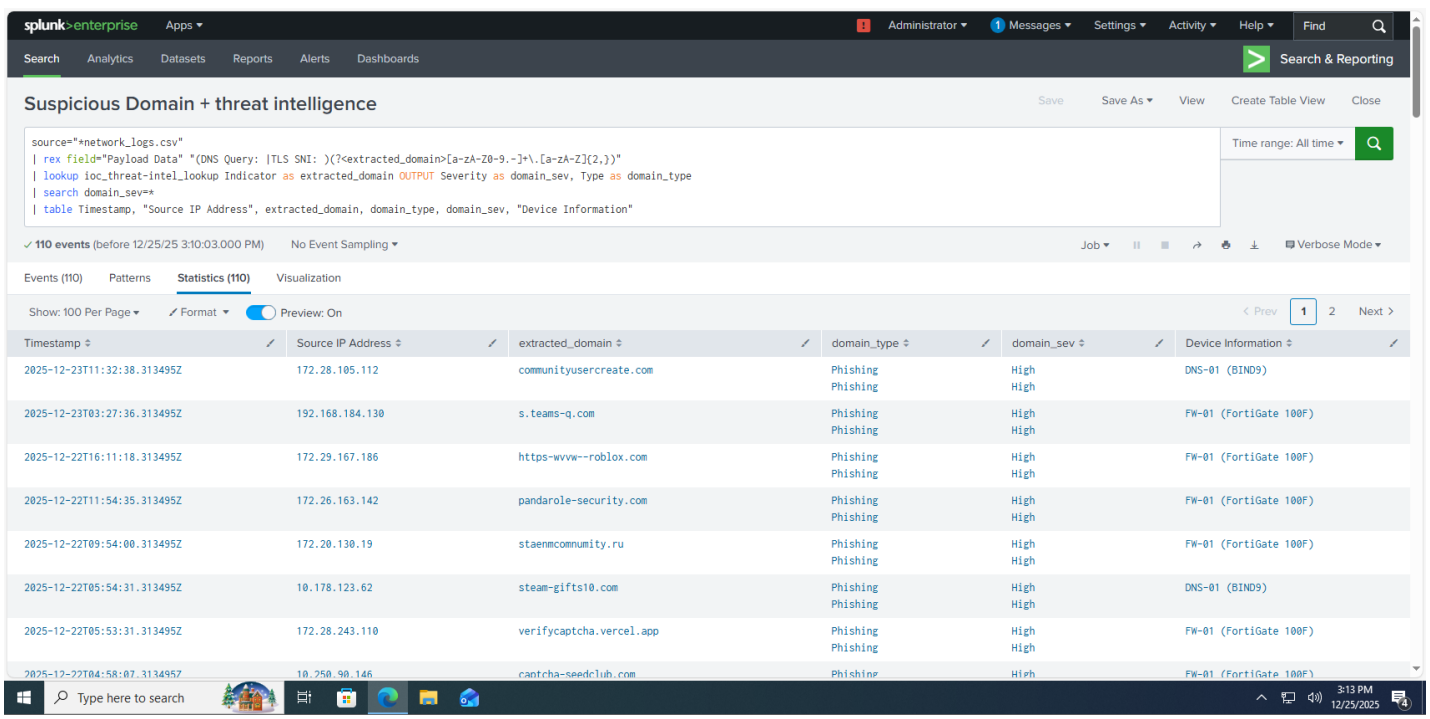
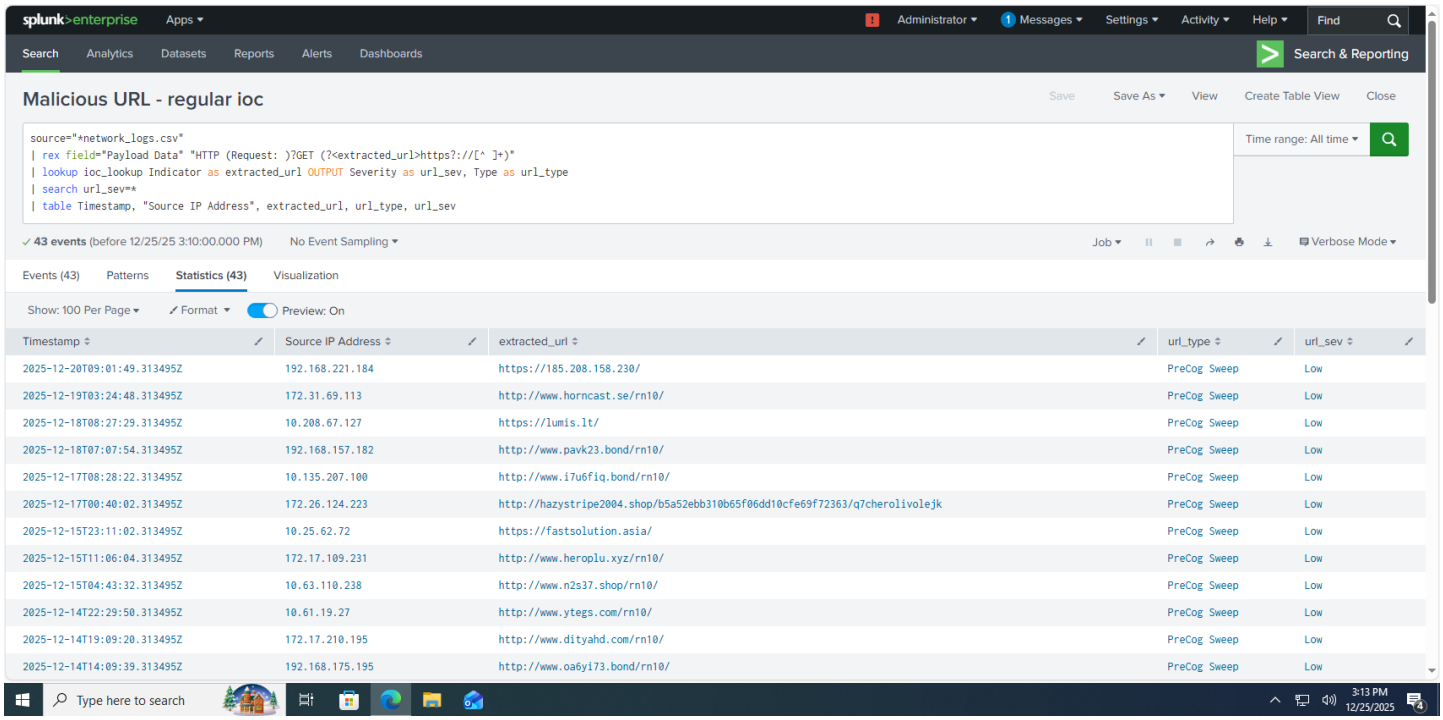


Figure 8 Domain correlation rules

- **URL rules:** matched URLs appearing in web/proxy activity.



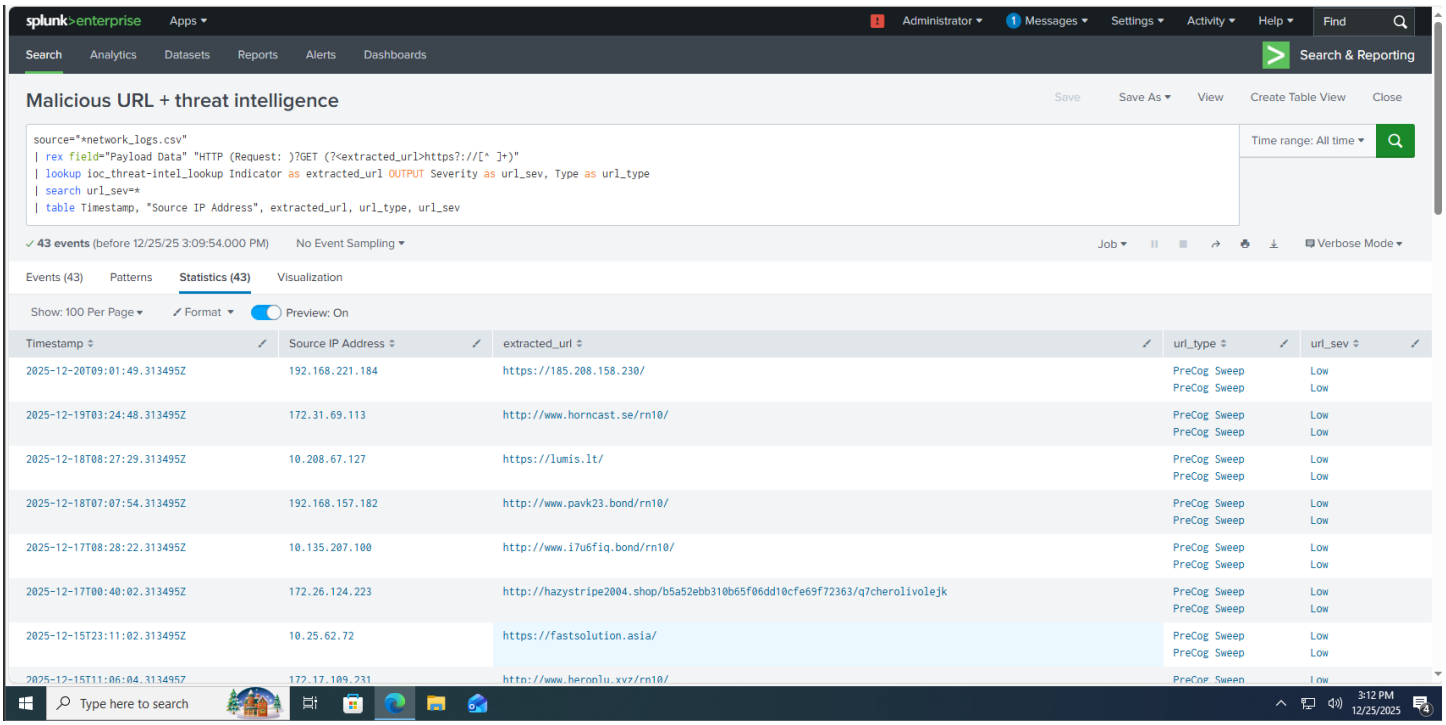
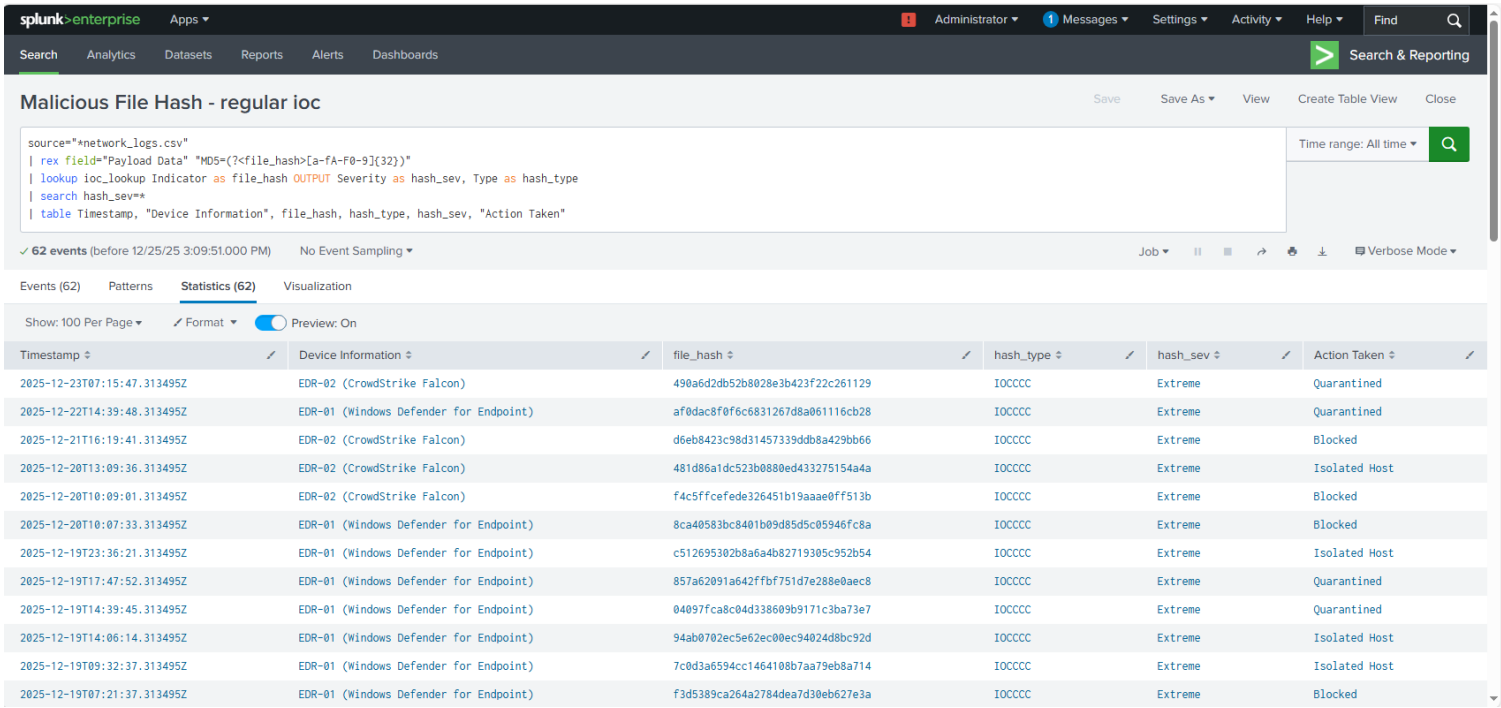


Figure 9 URL correlation rules

- **Hash rules:** matched hashes appearing in endpoint/EDR-style events (simulated malware execution/detection).



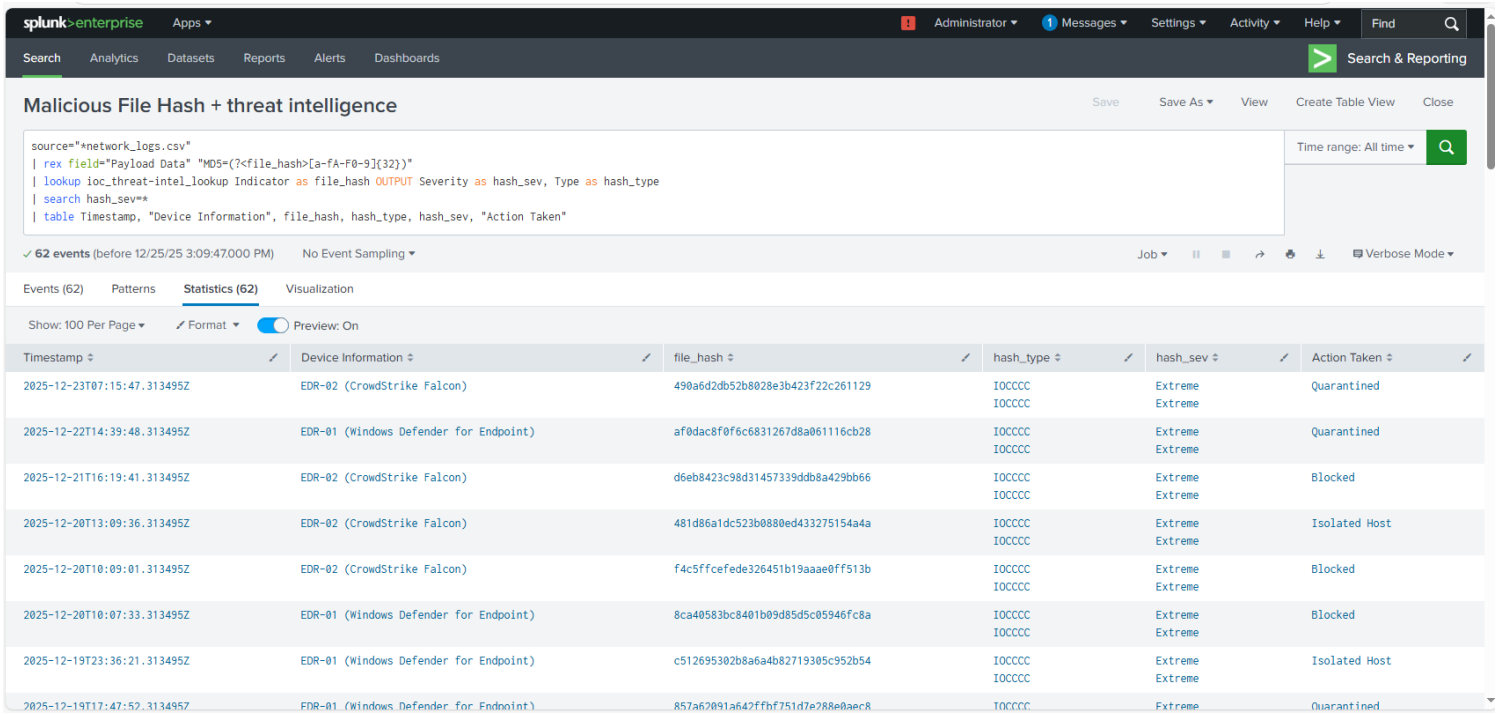
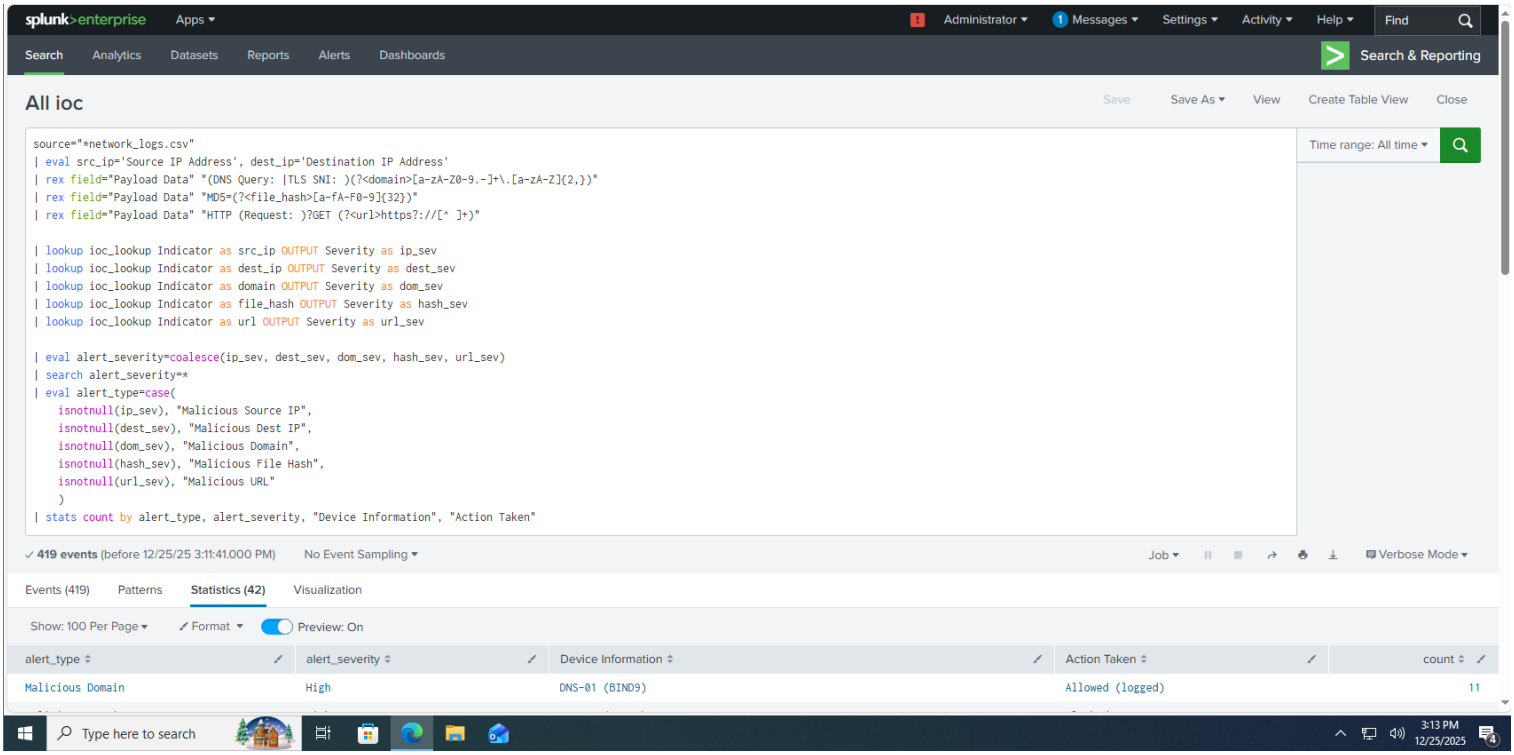


Figure 10 Hash correlation rule

- **All rules:** A combination of all the rules given above for an overview of all the system's threats.



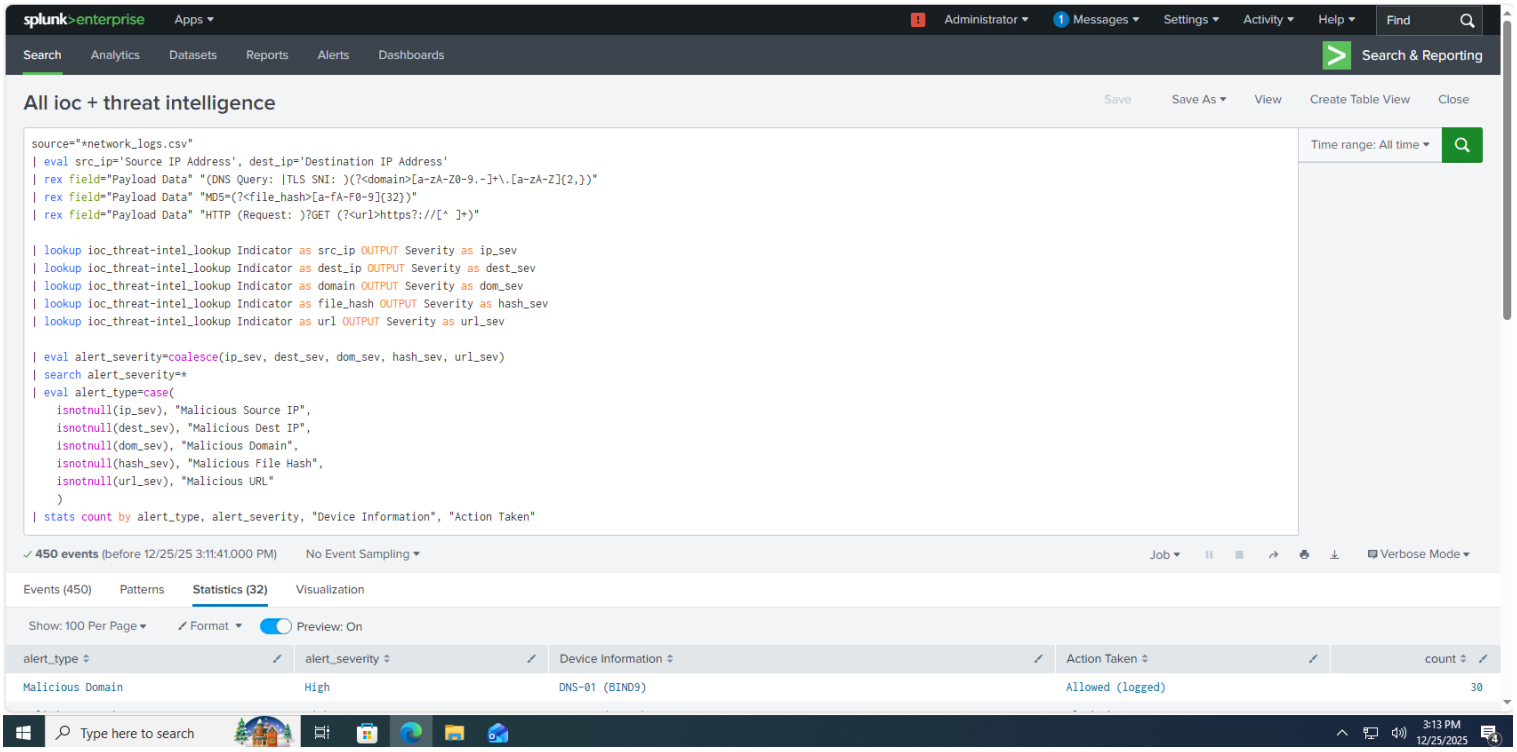


Figure 11 All correlation rules

7. Performance comparison on our simulated dataset

Our generated dataset contained 450 malicious IOC-matching events (embedded intentionally for testing).

Detection results by IOC type

IOC type	Malicious logs in dataset	Detected by Regular IOC	Detection rate	Detected by AI-enhanced IOC	Detection rate	Improvement
IPv4	235	205	87.23%	235	100%	**+30**
Domain	80	79	98.75%	80	100%	**+1**
Hostname	30	30	100%	30	100%	0
URL	43	43	100%	43	100%	0
File Hash	62	62	100%	62	100%	0
Total	**450**	**419**	**93.11%**	**450**	**100%**	**+31**

A. Interpretation (why AI-enhanced performed better)

The largest improvement was for IPv4 detection the AI-enhanced lookup contained many additional IP indicators that were not present in the regular list, so the AI-enhanced correlation rule detected 30 more malicious IP events.

Domain detection improved slightly (+1) while URL and hash detection stayed the same because both lookups contained the same unique URLs and hashes in our dataset.

Overall, the AI-enhanced approach achieved full coverage (100%) on the malicious test events, while the regular IOC lookup missed 31 events.

Quality and coverage of threat intelligence directly affect SIEM detection performance. A larger enriched IOC dataset can detect more malicious activity, especially for attacker infrastructure like IP ranges that change frequently.

8. How the system detects new events (continuous monitoring concept)

Once logs are ingested into Splunk, every new event is stored in the chosen index. Instead of running searches manually each time, we converted our correlation searches into scheduled alerts this makes detection automatic

- Splunk regularly searches the most recent events a rolling time window.
- It compares event fields against IOC lookup tables.
- When a match occurs, it generates an alert record with evidence.

This approach simulates real-time monitoring without needing advanced streaming systems.

A. Scheduling correlation rules every 15 minutes
To ensure frequent detection, each correlation rule was configured with:

This means the system repeatedly checks the newest data and quickly identifies suspicious activity.

Why 15 minutes?

- It is fast enough for lab detection and SOC-style monitoring.
- It reduces compute load compared to running every minute.
- It matches typical SOC alert schedules used in many environments.

B. Saving alerts for investigation (SOC evidence retention)
When a match occurs, Splunk stores the alert information so it can be investigated later. This is important because analysts need evidence even after the original event scrolls out of view.

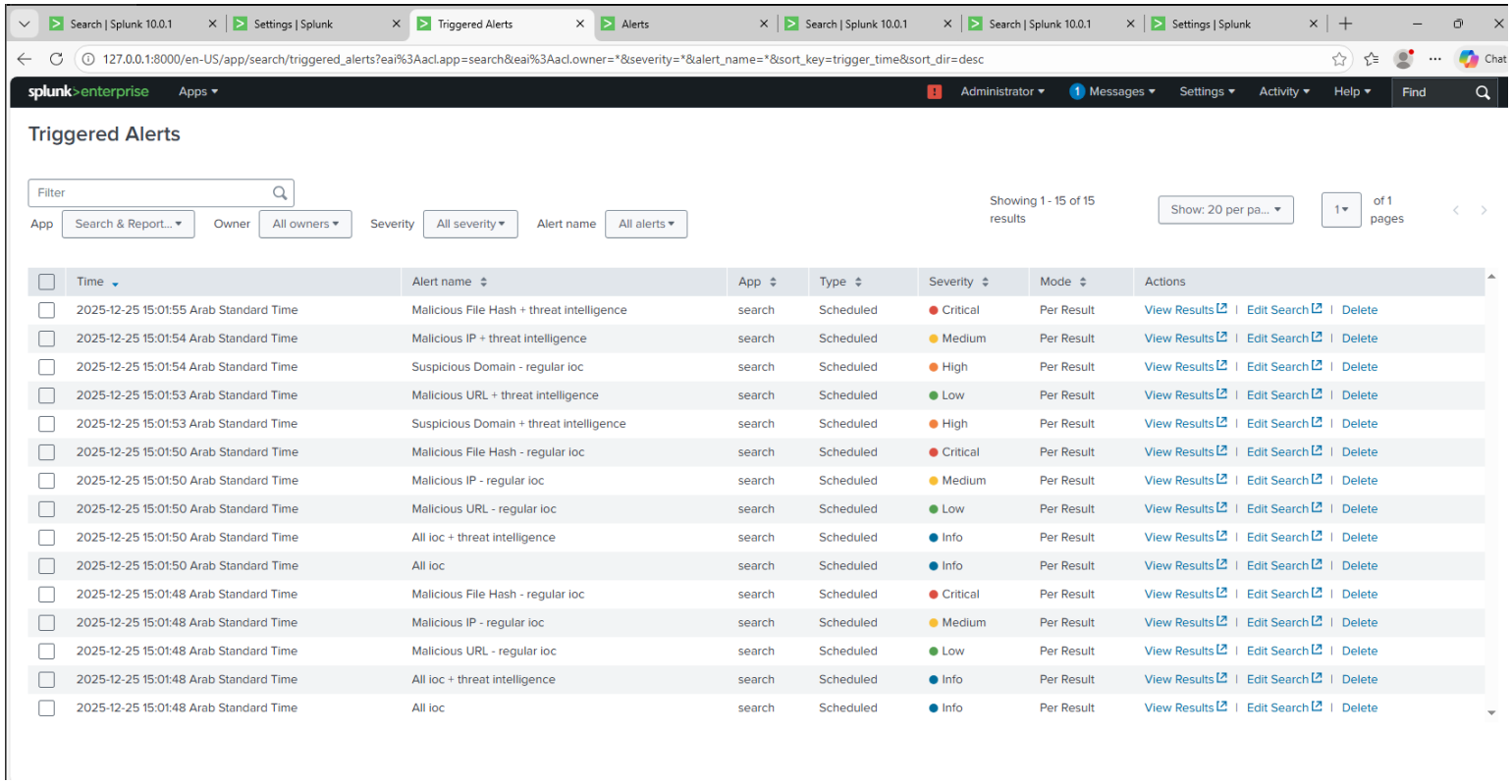


Figure 12 Saved Trigger Automatic Alerts

9. Conclusion

This project successfully implemented a complete beginner-friendly workflow for Threat Intelligence Integration in Splunk SIEM starting from external threat intelligence collection and ending with automated detection and alerting. We gathered a large IOC dataset from AlienVault OTX using an automated script then refined and organized the indicators into usable categories (IPv4, domains/hostnames, URLs, and file hashes). To strengthen detection coverage, we also applied an AI-assisted enrichment approach to identify related indicators and expand the IOC set producing an enhanced lookup that better represents attacker infrastructure.

To validate the solution in a safe lab environment we generated realistic simulated security logs that include both normal activity and intentionally injected IOC-matching malicious events. These logs were then ingested into Splunk and tested against IOC lookup tables using correlation searches. Separate correlation rules were created for each IOC type, and each rule was implemented in two versions regular IOC and AI-enhanced IOC to compare detection capability. The results showed that enrichment improves detection coverage with the most noticeable improvement occurring in IPv4-based detections due to the larger and more complete AI-enhanced indicator set.

Finally, the system was configured to operate continuously like a real SOC workflow correlation rules run automatically every 15 minutes triggering alerts whenever IOC matches occur. Alerts are saved for later reviews, allowing analysts to investigate evidence, understand the matched indicator and trace the related event context. Overall, the project demonstrates how integrating threat intelligence into a SIEM can significantly improve visibility and detection capability and it provides a practical foundation for future enhancements.

10. References

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